

Control the windmill tower direction

Equipment

TwinCAT 3 and the simulation "Twincat project WindMill"

Preparation

- Download the project "Twincat project WindMill" from Canvas to your computer
- Click on the file "TwinCAT Project Windmill.sln" to launch the project
- Start and select run <UmRT_Default>
- Click on "Activate Configuration"  and then answer OK to download the configuration to your runtime PLC.

Programming Tasks

Task 1 for grade G

- Use FBD and follow a good structure and naming standard.
- Create code that turns the windmill clockwise by ixRunCW, assigning qxRunCW, and counterclockwise, ixRunCCW, assigning qxRunCCW, when the selector is in manual mode (ixSelectorInAuto is false).
- In Auto mode, the yellow auto lamp qxInAutoMode should be on when the ixSelectorInAuto is TRUE, and the button ixButtonStartAuto is pressed. The lamp should be turned off when the Auto Manual selector turns to manual. Use self-holding or the RS function block for this, no other solutions are acceptable.
- In automode, use the in-signals ixPosDeviation and ixNegDeviation to control the outputs qxRunCW and qxRunCCW. The windmill should now turn slowly against the wind when the wind direction changes by clicking the wind direction buttons.
- The manual function should still work when the selector is set to manual.

Task 2 for grade VG

- Add a new POU **function**, give it the name **P**, Select Structure Text (ST) as the implementation language, and return type REAL.
- The input pins to the P-controller function should be named: SetValue, IsValue, and Kp. All in- pins should be REAL. Implement the formula for a P controller: $P := (SetValue - IsValue) * Kp$.

In FBD:

- Implement your new P controller function.
- Assign 0.0 to the in pin SetValue of the P controller, meaning no offset to the wind.
- Connect irActWindDev to the in pin IsValue of P controller.
- Assign 10.0 to the Kp.
- Use the CT, LT compare functions on the P controller output value to assign the qxRunCW and qxRunCCW.
- It should be possible to run both in On/Off mode using the code for task 1, running with the P-controller function, and Manual mode, selectable by ixSelectorPID and qxInAutoMode.
- Use the ABS function block on the P controller value to assign a positive real value to qrSetSpeed. The ABS function will convert negative values to positive values.
- Adjust Kp for good control. The windmill should not turn too slowly and should not overturn.

When you are done

- Show and explain your solution to the teacher. Save and upload your solution in case of.